

Programming Assignment 1

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****Course:**** CSEN 317 Distributed Systems

Project Overview

This project delivers a lightweight HTTP 1.0 web server written in Java. The server starts from the command line with two mandatory flags:

-document_root <path> - directory that holds static files

-port <8000-9999> - TCP port to listen on

At launch, the program automatically downloads the SCU home page and related assets into the document root, so `http://localhost:<port>/` is immediately reachable. The server is intended for local testing.

Project Structure

PA1/

```
|— src/main/java
|   |— WebServer.java
|   |— WebMethod.java
|— documents/
|— Readme.pdf
```

Run the Project

Execute these orders at root directory of the project

1. compile java source file

```
javac -d . src/main/java/WebServer.java src/main/java/WebMethod.java
```

2. Launch the server (port 8080, directory ./documents)

```
java WebServer -document_root ./documents -port 8080
```

Screenshot of Connecting to the Server

```
PS C:\Users\pm\Desktop\PA1> curl -v http://localhost:8080/index.html
VERBOSE: GET http://localhost:8080/index.html with 0-byte payload
VERBOSE: received 79310-byte response of content type text/html

StatusCode      : 200
StatusDescription : OK
Content         : <!DOCTYPE html>
                  <html lang="en">
                  <head>
                    <meta charset="utf-8">
                    <meta http-equiv="X-UA-Compatible" content="IE=edge">
                    <meta name="viewport" content="width=device-width, initial-scale=1">
                  ...
RawContent      : HTTP/1.0 200 OK
                  Content-Length: 79310
                  Content-Type: text/html

                  <!DOCTYPE html>
                  <html lang="en">
                  <head>
                    <meta charset="utf-8">
                    <meta http-equiv="X-UA-Compatible" content="IE=edge">
                    <meta...
Forms           : {, }
Headers         : {[Content-Length, 79310], [Content-Type, text/html]}
Images          : [{@{innerHTML=; innerText=; outerHTML=<img class="vv-thumbnail active video component" id="thumbnail_1" draggable="false" alt="Sam Bertram '16, M.S. '18, with his arms crossed proudly, in front of his vertical farm" src="https://www.scu.edu/media/external/brand-rollout-announcement/sam-thumbnail-still-02.jpg"; outerText=; tagName=IMG; class=vv-thumbnail active video component; id=thumbnail_1; draggable=false; alt=Sam Bertram '16, M.S. '18, with his arms crossed proudly, in front of his vertical farm" src="https://www.scu.edu/media/external/brand-rollout-announcement/sam-thumbnail-still-02.jpg"}, @{innerHTML=; innerText=; outerHTML=<img class="vv-thumbnail active video component" id="thumbnail_2" draggable="false" alt="Mirelle Raza, a Santa Clara Graduate, smiling while holding a book and grasping her blazer in a library" src="https://www.scu.edu/media/external/brand-rollout-announcement/marielle-thumbnail-still-02.jpg"; outerText=; tagName=IMG; class=vv-thumbnail active video component; id=thumbnail_2; draggable=false; alt=Mirelle Raza, a Santa Clara Graduate, smiling while holding a book and grasping her blazer in a library" src="https://www.scu.edu/media/external/brand-rollout-announcement/marielle-thumbnail-still-02.jpg"}, @{innerHTML=; innerText=; outerHTML=<img class="sun-small" alt="small red sun icon" src="https://www.scu.edu/media/external/brand-rollout-announcement/sun-red.svg"; outerText=; tagName=IMG; class=sun-small; alt=small red sun icon" src="https://www.scu.edu/media/external/brand-rollout-announcement/sun-red.svg"}, @{innerHTML=; innerText=; outerHTML=<img class="hp-story-img w-100 h-auto img-fluid lazysrcset" alt="Seeking Humanity in Healthcare" src="https://upload.wikimedia.org/wikipedia/commons/c/ca/1x1.png" data-3="/media/institutional-pages/news-amp-events/Sydney-Shelby-1160x733-360x227.gif" data-2="/media/institutional-pages/news-amp-events/Sydney-Shelby-1160x733-360x227.gif" data-1="/media/institutional-pages/news-amp-events/Sydney-Shelby-1160x733-360x227.gif"}]}
```

Screenshot of visiting localhost

